

Varsha Thiyagarajan

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Education

B.Tech in Artificial Intelligence & Machine Learning 2024 – 2028
Sri Ramachandra Faculty of Engineering and Technology, Chennai
CGPA: 9.16
Relevant Coursework: Machine Learning, Data Structures, Object-Oriented Programming, DBMS, Operating Systems

Skills

- **Programming Languages:** Python, Java, JavaScript
- **AI / Machine Learning:** Scikit-learn, NumPy, Pandas, Matplotlib
- **Frameworks:** React.js, Next.js, Node.js, Express.js, Flask
- **Databases:** MongoDB, MySQL
- **Tools:** Git, GitHub, Postman, Power BI, VS Code, Overleaf
- **Technologies:** REST APIs, Authentication Systems, Data Visualization, Responsive Web Design

Projects

Learning Style Classifier using VAK Model (EdTech + ML) Python, Scikit-learn, Flask
GitHub Repository

- Developed an end-to-end ML system to classify learners into Visual, Auditory, or Kinesthetic styles, improving classification accuracy through optimized feature engineering and preprocessing.
- Built a scalable ML pipeline handling data preprocessing, feature extraction, model training, and API-based prediction for real-time inference.
- Deployed classification models via a Flask backend with an interactive UI, enabling seamless real-time learning style predictions.

FlashMind — AI-Powered Intelligent Tutoring System Python, LLM APIs, Flask, React
GitHub Repository

- Engineered an AI tutoring platform that transforms unstructured study materials into structured summaries, quizzes, and flashcards, improving content accessibility and learning efficiency.
- Built automated OCR and PDF parsing pipelines using PyPDF2 to process and extract structured knowledge from diverse learning documents.
- Integrated source-grounded LLM generation to deliver adaptive and personalized learning experiences based on user interactions.

AI-Driven Health Prediction & Medical Report Generator Python, Scikit-learn, Flask
GitHub Repository

- Developed a machine learning-based healthcare assistant to predict potential health risks using classification models trained on structured medical data.
- Built an NLP-based report generation system to convert predictions into clear, patient-friendly insights, improving interpretability of results.
- Designed a Flask backend to integrate prediction models with dynamic report generation for real-time usage.

MeetSphere — Full Stack Event Management Platform MERN Stack
GitHub Repository

- Developed a full-stack event platform supporting event creation, user management, and role-based access control with scalable architecture.
- Implemented secure authentication using JWT and built RESTful APIs with Node.js and Express.js to handle user and event workflows.
- Designed reusable React components and integrated MongoDB-backed services to efficiently manage event lifecycle operations.

Achievements

- Winner — Power BI Hackathon for developing a disaster rescue resource optimization dashboard. View Details

Certifications

- Data Structures and Algorithms — GeeksforGeeks Credential
- Full Stack Development — Udemy Credential
- MongoDB Developer Badge — MongoDB University Credential
- Machine Learning for AI and Data Scientist — Code Basics Credential